OUTLINE

01 *Dataset*

02 *Loudness and explicity throughout the years*

03 *Evolution of music*

04 *Data insights on artists*

Mini Project

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2:54

3:49

Dataset

1. Includes 174,170 spotify songs
2. Range: from 1921 until 2012
3. Attributes of the songs: acousticness, artists, danceability, duration_ms, energy, explicit, id, instrumentalness, key, liveness, loudness, mode, name, popularity, release_date, speechiness, tempo, valence, year

spotify

Filter

	acousticness	artists	danceability	du
1	0.9910	['Mamie Smith']	0.5980	
2	0.9930	['Mamie Smith']	0.6470	
3	0.9960	['Mamie Smith & Her Jazz Hounds']	0.4240	
4	0.9920	['Mamie Smith']	0.7820	
5	0.9960	['Mamie Smith & Her Jazz Hounds']	0.4740	
6	0.9520	['Dorville']	0.6880	
7	0.9840	['Dick Haymes']	0.2690	
8	0.3810	['ST']	0.9360	
9	0.9440	['Félix Mayol']	0.6560	
10	0.9960	['Maurice Chevalier']	0.6130	

Showing 1 to 10 of 174,170 entries, 19 total columns

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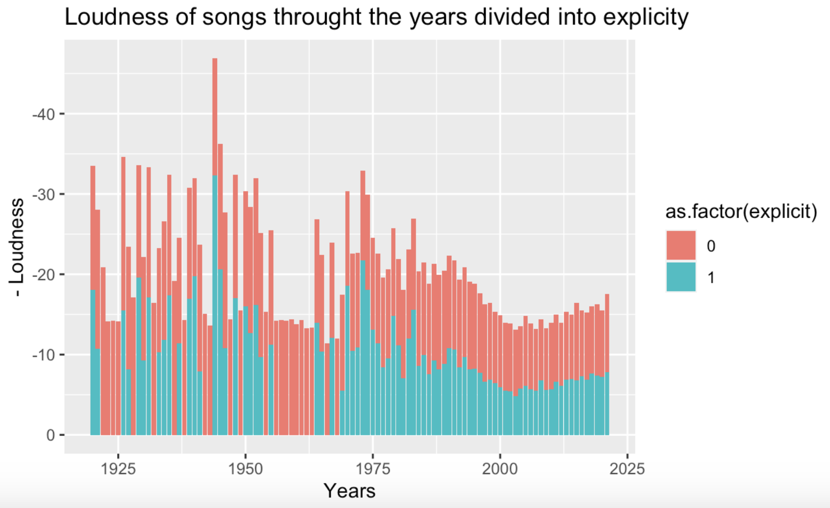
2:54

3:49

How did the loudness change throughout the years? Is it more apparent for songs containing explicit content?

My hypothesis: Loudness of songs increase throughout the years, but explicit songs are louder than non-explicit ones.



This hypothesis was proven wrong, since according to the graph, the non-explicit songs tend to be louder than the explicit ones and the loudness of songs also decrease which is unexpected.

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2:54
3:49

The most popular songs

Assumption: Since English is understood and spoken by the majority of people, I assume that that would be the language of all the top songs.

However, shown in this popularity chart, amongst the top 12 songs, there are also 3 Spanish songs apparent.

Top 12 most popular songs lately:

	artists	name
1	['Olivia Rodrigo']	drivers license
2	['24kGoldn', 'iann dior']	Mood (feat. iann dior)
3	['Ariana Grande']	positions
4	['Bad Bunny', 'Jhay Cortez']	DÁKITI
5	['KAROL G']	BICHOTA
6	['Ariana Grande']	34+35
7	['CJ']	Whoopty
8	['The Kid LAROI']	WITHOUT YOU
9	['Billie Eilish']	Therefore I Am
10	['Bad Bunny', 'ROSALÍA']	LA NOCHE DE ANOCHE
11	['Pop Smoke']	What You Know Bout Love
12	['Tate McRae']	you broke me first

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2:54

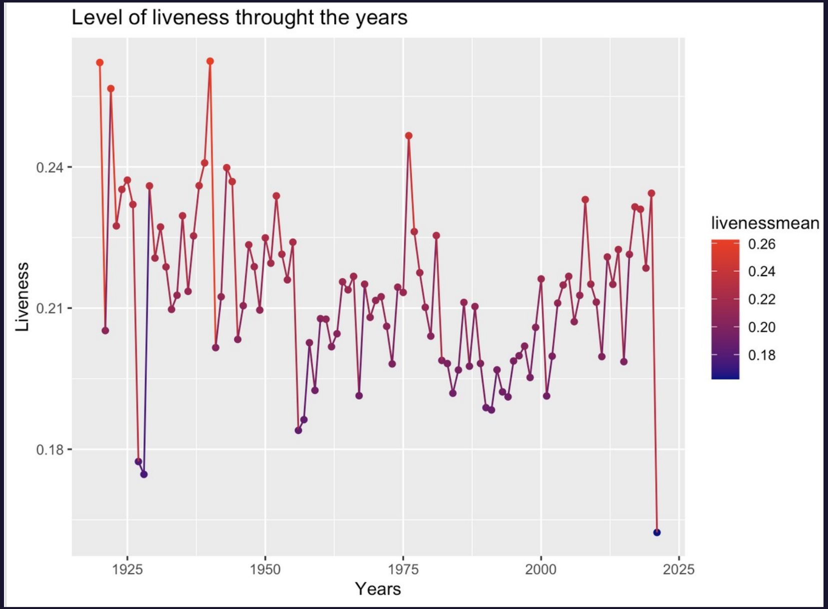
3:49

Evolution of music

Liveness: The relative duration of the track sounding as a live performance

Hypthesis: I expect the level of liveness throught the years to decrease, especially around the years 2010-2021, because with autotune and new technology, music can be created with even artificial intelligence.

As seen in the graph, this is exactly how it happened.



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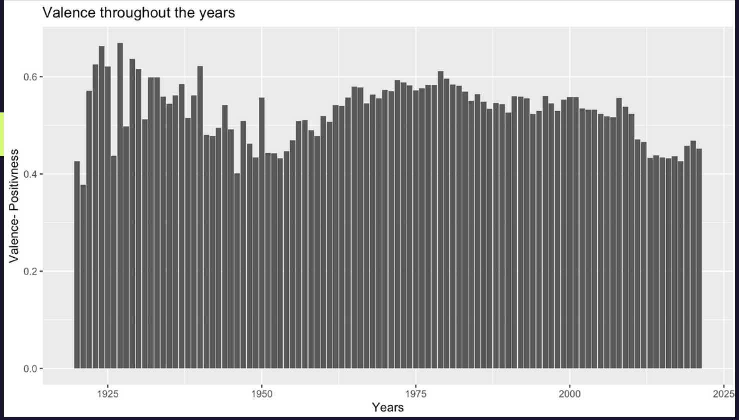
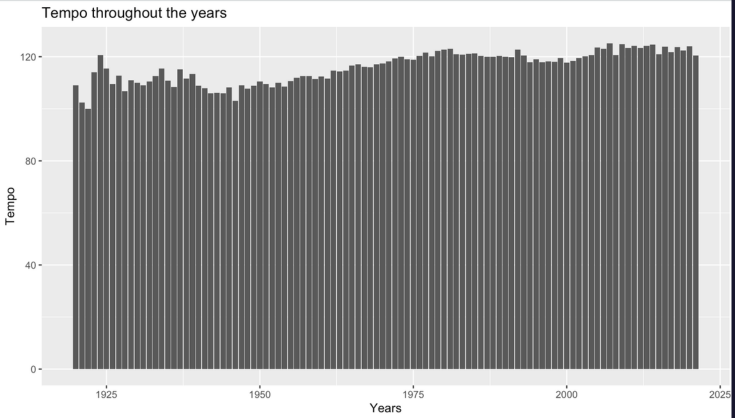
2:54
3:49

The Great Depression

Hypothesis: During the Great Depression the tempo and the valence of songs decreased due to the hard times the United States had to go through.

Even though according to the results, the tempo was indeed slower during the Great Depression, the valence of songs was actually higher. That could be the reason for artists trying to help the country by releasing more positive songs in order to make people have a better mood.





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2:54

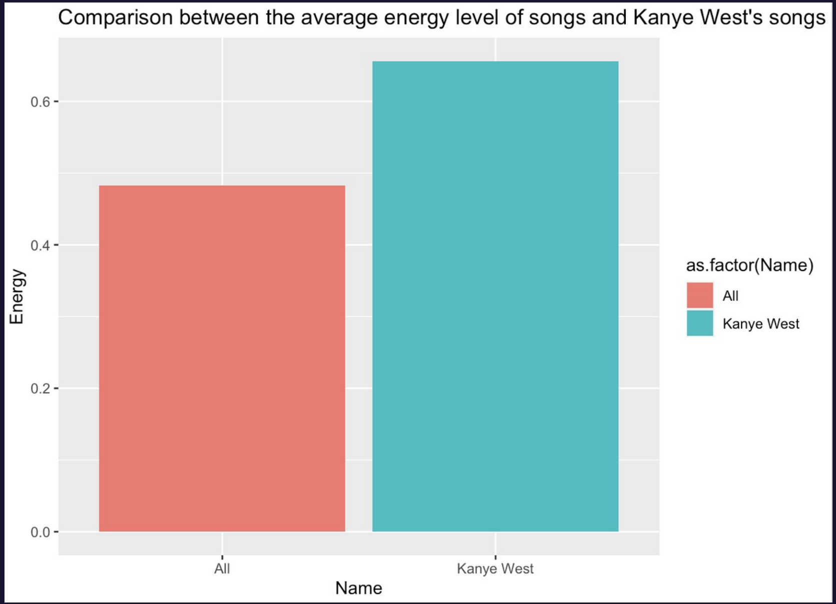
3:49

Data insights on artists- Kanye West

Kanye West is a very well-known rapper, who has a very wide range of popular songs in the genre of hip-hop as well.

Since he is a rapper, it is expected for his songs to be more energetic than songs in general.

By filtering his songs and finding their average energy level, the graph displays accurately the difference between his song and every other song.



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3:49

Code for each of the segments

Reading the csv file and loading all necessary libraries

```
library(readr)
df <- read_csv("Desktop/Data Visualization/spotify.csv")
glimpse(df)
View(df)
library(dplyr)
library(ggplot2)
```

Loudness of songs throught the years and the ratio between explicit and non-explicit songs

```
ggplot(df) +
  geom_bar(aes(x=year , y=loudness , fill=as.factor(explicit)),stat="summary") +
  scale_y_reverse() +
  labs(title="Loudness of songs throught the years divided into explicity") +
  xlab("Years") +
  ylab("- Loudness")
```

Top 12 most popular songs lately:

```
library(dplyr)
top12<-df %>% arrange(desc(popularity)) %>% select(artists,name) %>% head(12)
View(top12)
```

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3:49

Code for each of the segments

Level of liveness throughout the years

```
newdf<-df %>%
  group_by(year) %>%
  summarise(livenessmean=mean(liveness))  #group the liveness level by years

ggplot(newdf, aes(x=year,y=livenessmean, color=livenessmean)) +
  geom_point() +
  geom_line() +
  ylab("Liveness") +
  xlab("Years") +
  labs(title="Level of liveness throught the years") +
  scale_color_gradient(low="darkblue",high="red")
```

Valence and Tempo in the Great depression

```
  #Valence
ggplot(df) + geom_bar(aes(x=year, y=valence),stat="summary") +
  ylab("Valence- Positivness") +
  xlab("Years") +
  labs(title="Valence throughout the years")

  #Tempo
ggplot(df) + geom_bar(aes(x=year, y=tempo),stat="summary") +
  ylab("Tempo") +
  xlab("Years") +
  labs(title="Tempo throughout the years")
```

Songs of Kanye West

```
df$artists <- lapply(df$artists, "as.character")  # Convert from factor variable to string
kanye<-df %>% filter(grepl("Kanye West", artists, fixed=TRUE))  # Search the artists by substring
View(kanye)

energyall<-df %>% pull(energy) %>% mean()   #finding the average energy of all songs
kanyeenergy<- kanye %>% pull(energy) %>% mean()  #finding the average energy of kanye song

original<-data.frame(c("All", "Kanye West"),c(energyall,kanyeenergy))

colnames(original)[colnames(original)=="c.energyall..kanyeenergy."] <- "Energy"
colnames(original)[colnames(original)=="c..All....Kanye.West.."] <- "Name"

ggplot(original) +
  geom_bar(aes(x=Name, y=Energy, fill=as.factor(Name)), stat="summary") +
  labs(title="Comparison between the average energy level of songs and Kanye West's songs")
```

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2:54

3:49